Started on	Wednesday, 9 September 2026, 10:54 PM
State	Finished
Completed on	Wednesday, 9 September 2026, 11:10 PM
Time taken	15 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int cmp(const void *a,const void *b){
    return *(int *)a - *(int *)b;
}
int main(){
    int n,a[1000],sum=0;
    scanf("%d",&n);
    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);
    qsort(a,n,sizeof(int),cmp);
    for(int i =0;i<n;i++)
        sum+=a[i]*i;
    printf("%d",sum);
    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.